

IaaS, PaaS, SaaS – Detailed Explanation

IaaS (Infrastructure as a Service)

IaaS is a cloud service model that provides **basic computing resources** such as virtual machines, storage, and networking over the internet. In this model, the cloud provider manages the physical hardware, while the user is responsible for managing the operating system, applications, and data.

It gives the highest level of control among the three models. Users can configure and manage their systems as needed, just like working on a physical computer.

Example: Amazon Web Services (AWS EC2)

AWS EC2 allows users to rent virtual servers. Users can install their own operating system, run applications, and store data. The provider handles the infrastructure, but the user controls everything else.

Best suited for: Developers, system administrators, and organizations needing full control over their environment.

PaaS (Platform as a Service)

PaaS provides a **complete platform for application development and deployment**. The cloud provider manages the infrastructure, operating system, and runtime environment. Users only need to focus on writing and managing their application code.

This reduces the complexity of managing servers and allows developers to build applications quickly.

Example: Microsoft Azure App Services

Azure App Services allows developers to build, test, and deploy web applications without

worrying about infrastructure or system maintenance. It also provides automatic scaling and updates.

Best suited for: Application developers and startups who want to focus on coding rather than system management.

SaaS (Software as a Service)

SaaS is a cloud service model that delivers **ready-to-use software applications over the internet**. Users do not need to install or maintain the software. Everything is managed by the service provider.

Users can access the application through a web browser or mobile app.

Example: Gmail

Gmail is an email service where users can send and receive emails without installing any software. All data storage, security, and updates are handled by the provider.

Best suited for: End users and businesses who need easy access to software without technical setup.

Comparison Summary

- **IaaS:** Maximum control, user manages OS and applications
- **PaaS:** Medium control, user focuses only on application development
- **SaaS:** Minimum control, user simply uses the software